

Yuzheng Hu

✉ YH46@ILLINOIS.EDU 🏠 mirnegg.github.io

EDUCATION	University of Illinois at Urbana-Champaign <i>Ph.D. in Computer Science</i> ◦ Advisor: Han Zhao Peking University <i>B.S. in Mathematics (Math and Applied Math, Honor Track)</i> ◦ Advisors: Wei Lin and Liwei Wang	Urbana, IL, USA <i>Aug. 2021 – May. 2026 (expected)</i> Beijing, China <i>Sep. 2017 – Jul. 2021</i>
RESEARCH INTERESTS	I develop the science of data —a perspective that views data as an <i>optimizable</i> , <i>expandable</i> and <i>predictable</i> component of AI systems. To this end, I work on data attribution , synthetic data , and privacy .	
PUBLICATIONS	★ stands for equal contribution or alphabetical order	
	[P1] A benchmark of expert-level academic questions to assess AI capabilities 📄 🌐 📊 Dataset contributor of Humanity’s Last Exam; authored 4 of 2,500 expert-level questions, with one selected as a prize-winning entry (top 550 worldwide) <i>Nature</i> (2026)	
	[P2] A Snapshot of Influence: A Local Data Attribution Framework for Online Reinforcement Learning 📄 🔄 Yuzheng Hu* , Fan Wu*, Haotian Ye, David Forsyth, James Zou, Nan Jiang, Jiaqi W. Ma, Han Zhao <i>The 39th Annual Conference on Neural Information Processing Systems (NeurIPS 2025, Oral)</i>	
	[P3] Taming Hyperparameter Sensitivity in Data Attribution: Practical Selection Without Costly Retraining 📄 Weiyi Wang, Junwei Deng, Yuzheng Hu , Shiyuan Zhang, Xirui Jiang, Runting Zhang, Han Zhao, Jiaqi W. Ma <i>The 39th Annual Conference on Neural Information Processing Systems (NeurIPS 2025)</i>	
	[P4] MergeBench: A Benchmark for Merging Domain-Specialized LLMs 📄 🔄 Yifei He, Siqi Zeng, Yuzheng Hu , Rui Yang, Tong Zhang, Han Zhao <i>The 39th Annual Conference on Neural Information Processing Systems (NeurIPS 2025, Datasets and Benchmarks Track)</i>	
	[P5] Empirical Privacy Variance 📄 🔄 Yuzheng Hu* , Fan Wu*, Ruicheng Xian, Yuhang Liu, Lydia Zakynthinou, Pritish Kamath, Chiyuan Zhang, David Forsyth <i>The 42nd International Conference on Machine Learning (ICML 2025)</i>	
	[P6] An Improved Autoregressive Evaluation Paradigm for Large Language Models 📄 Jipeng Zhang*, Rui Pan*, Yuzheng Hu* , Kashun Shum*, Guanyu Yao, Xiang Liu, Renjie Pi, Hanze Dong, Shizhe Diao, Yong Lin, Han Zhao, Tong Zhang <i>ACM Transactions on Intelligent Systems and Technology (ACM TIST 2025)</i>	
	[P7] Most Influential Subset Selection: Challenges, Promises, and Beyond 📄 🔄 Yuzheng Hu , Pingbang Hu, Han Zhao, Jiaqi W. Ma <i>The 38th Annual Conference on Neural Information Processing Systems (NeurIPS 2024)</i>	
	[P8] Localize-and-Stitch: Efficient Model Merging via Sparse Task Arithmetic 📄 🔄 Yifei He, Yuzheng Hu , Yong Lin, Tong Zhang, Han Zhao <i>Transactions on Machine Learning Research (TMLR 2024)</i>	
	[P9] SoK: Privacy-Preserving Data Synthesis 📄 🌐 Yuzheng Hu* , Fan Wu*, Qinbin Li, Yunhui Long, Gonzalo Munilla Garrido, Chang Ge, Bolin Ding,	

David Forsyth, Bo Li, Dawn Song

The 45th IEEE Symposium on Security and Privacy (S&P 2024)

- [P10] HPL-ViT: A Unified Perception Framework for Heterogeneous Parallel LiDARs in V2V 
Yuhang Liu, Boyi Sun, Yuke Li, **Yuzheng Hu**, Fei-Yue Wang
The 41st IEEE International Conference on Robotics and Automation (ICRA 2024)
- [P11] Revisiting Scalarization in Multi-Task Learning: A Theoretical Perspective 
Yuzheng Hu, Ruicheng Xian, Qilong Wu, Qiuling Fan, Lang Yin, Han Zhao
The 37th Annual Conference on Neural Information Processing Systems (NeurIPS 2023)
- [P12] Understanding the Impact of Adversarial Robustness on Accuracy Disparity  
Yuzheng Hu, Fan Wu, Hongyang Zhang, Han Zhao
The 40th International Conference on Machine Learning (ICML 2023)
- [P13] Actor-critic is implicitly biased towards high entropy optimal policies 
Yuzheng Hu*, Ziwei Ji*, Matus Telgarsky*
The 10th International Conference on Learning Representations (ICLR 2022)
- [P14] Towards Understanding the Data Dependency of Mixup-style Training  
Muthu Chidambaram, Xiang Wang, **Yuzheng Hu**, Chenwei Wu, Rong Ge
*The 10th International Conference on Learning Representations (ICLR 2022, **Spotlight**)*
- [P15] Don't Waste Your Bits! Squeeze Activations and Gradients for Deep Neural Networks via TinyScript 
Fangcheng Fu, **Yuzheng Hu**, Yihan He, Jiawei Jiang, Yingxia Shao, Ce Zhang, Bin Cui
The 37th International Conference on Machine Learning (ICML 2020)

- MANUSCRIPTS [M1] ACTG-ARL: Differentially Private Conditional Text Generation with RL-Boosted Control  
Yuzheng Hu, Ryan McKenna, Da Yu, Shanshan Wu, Han Zhao, Zheng Xu, Peter Kairouz
- [M2] A Survey of Data Attribution: Methods, Applications, and Evaluation in the Era of Generative AI 
Junwei Deng*, **Yuzheng Hu***, Pingbang Hu*, Ting-Wei Li*, Shixuan Liu*, Jiachen T. Wang, 21 others, Jiaqi W. Ma

INVITED TALKS **Differentially Private Conditional Text Generation with RL-Boosted Control**

- Google Athena seminar Sep. 2025
- Google ML Privacy seminar (Host: Natalia Ponomareva) Sep. 2025

Empirical Privacy Variance

- Google ML Privacy seminar (Host: Matthew Jagielski) Mar. 2025

INTERNSHIP EXPERIENCE

Google Research

Mountain View, CA, US

Research Intern, Hosts: Zheng Xu, Ryan McKenna, Da Yu; Manager: Peter Kairouz

May. 2025 – Nov. 2025

- Developed a hybrid approach for DP synthetic text generation that advances the state of the art [M1]

Jane Street Group, LLC

New York, NY, US

Quantitative Trading Intern

May. 2024 – July. 2024

- Ranked 2/200 (1/150 among interns) in the Tenth Annual Figgie World Championships
- Participated in mock trading games and worked on two desk projects (options and equities)

Damo Academy, Alibaba Group (U.S.) Inc.

Bellevue, WA, US

Research Intern, Mentor: Bolin Ding

May. 2022 – Aug. 2022

	◦ Worked on the intersection of privacy and law ◦ Wrote a systematization of knowledge paper on privacy-preserving data synthesis [P9]	
RESEARCH EXPERIENCE	Simons Institute for the Theory of Computing  <i>Visiting Graduate Student, Program: Modern Paradigms in Generalization</i> Berkeley, CA, US Aug. 2024 – Dec. 2024 ◦ Quantified the discrepancy between differential privacy and memorization in LLMs [P5], with Pritish Kamath and Chiyuan Zhang University of Illinois at Urbana-Champaign <i>Graduate Research Assistant, Advisor: Han Zhao</i> Urbana, IL, US Feb. 2022 – now ◦ Thesis research: data attribution [P2] [P3] [P7] [M2], with Jiaqi Ma ◦ Other works: model merging [P4] [P8]; representational limitations of scalarization in multi-task learning [P11]; trade-off between robustness and fairness [P12] University of Illinois at Urbana-Champaign <i>Graduate Research Assistant, Advisor: Matus Telgarsky</i> Urbana, IL, US Aug. 2021 – Dec. 2021 ◦ Proved that actor-critic converges to high entropy optimal policy without explicit regularization [P13] Duke University <i>Undergraduate Research Assistant, Advisor: Rong Ge</i> Remote Jun. 2020 – Feb. 2021 ◦ Demonstrated the data-dependency of mixup [P14] School of Intelligence Science and Technology, Peking University <i>Undergraduate Research Assistant, Advisor: Liwei Wang</i> Beijing, China Aug. 2019 – Jun. 2020 ◦ Identified two non-local properties of the loss landscape of neural networks School of Computer Science, Peking University <i>Undergraduate Research Assistant, Advisor: Bin Cui</i> Beijing, China Aug. 2018 – Aug. 2019 ◦ Introduced a distribution-aware, non-uniform quantization algorithm for multilayer perceptrons [P15]	
TEACHING EXPERIENCE	UIUC CS 591 – Machine Learning Reading Group UIUC CS 598 – Transfer Learning UIUC CS 498 – Trustworthy Machine Learning	Spring 2024, Fall 2024 Spring 2023 Fall 2022
SERVICES	Seminar Co-Organizer: UIUC Machine Learning Seminar  Workshop Co-Organizer: DATA-FM @ ICLR 2026  Program Committee: TPDP (2025) , ICLR SynthData (2025) , ACL L2M2 (2025)  Reviewer: NeurIPS (2022-2025), ICML (2023-2025), ICLR (2024-2026), AISTATS (2024), ICCV (2025), TMLR	2024 – now
HONORS AND AWARDS	NeurIPS 2025 Oral (Top 0.3%, 77/21,575) NeurIPS 2025 Top Reviewer (Top 8%)  ICML 2025 Outstanding Reviewer (Top 2%)  Prize-Winning Question in Humanity's Last Exam (Top 1%, 550/70,000+, \$500)  NeurIPS 2023 Top Reviewer (Top 10%)  Saburo Muroga Endowed Fellowship (Top 4% in UIUC, 4/100+)  Exceptional Award for Academic Innovation (Top 1% in Peking University) Elite Undergraduate Training Program of Applied Math (Top 15% in SMS)  Invited Participant, China TST for the 58th International Mathematical Olympiad (IMO)	2025 2025 2025 2025 2023 2021-2022 2020 2019-2021 2017

	Gold Medal, 32nd Chinese Mathematical Olympiad (CMO) (Top 60 in China)	2016
	Scored 771/810 in Jr. High Entrance Exam (Top 0.01% in Guangzhou, 10/100,000+)	2014
TRAVEL GRANTS	TPDP Student Travel Funding (\$825)	2025
	IEEE S&P Student Travel Grants Award (\$1000)	2024
	NeurIPS 2023 Scholar Award (\$2000)	2023
SKILLS	Programming Python, Shell Script, $\text{\LaTeX}$, MATLAB Frameworks & Packages PyTorch, HuggingFace	